

Replication Report: Koll & Abbot (2016)

Decoupling Thermal Inversions and Atmospheric Circulation on Synchronous Exoplanets

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Abstract

This report details the full replication of Figures 1 and 2 from Koll & Abbot (2016) (*ApJ*, 825, 99) using our `hot_jupiter` C++ core library (`Koll2016InversionModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9999$) across day-night temperature difference ΔT_{dn} and thermal inversion strength η_{inv} vs shortwave-to-longwave opacity ratio γ .

1 Introduction & Methodology

Koll & Abbot (2016) introduce analytical scaling models for atmospheric circulation and thermal inversions:

$$\Delta T_{\text{dn}} = \frac{T_{\text{eq}}}{1 + \frac{\tau_{\text{rad}}}{\tau_{\text{adv}}}} \quad (1)$$

2 Replication Results & Figures

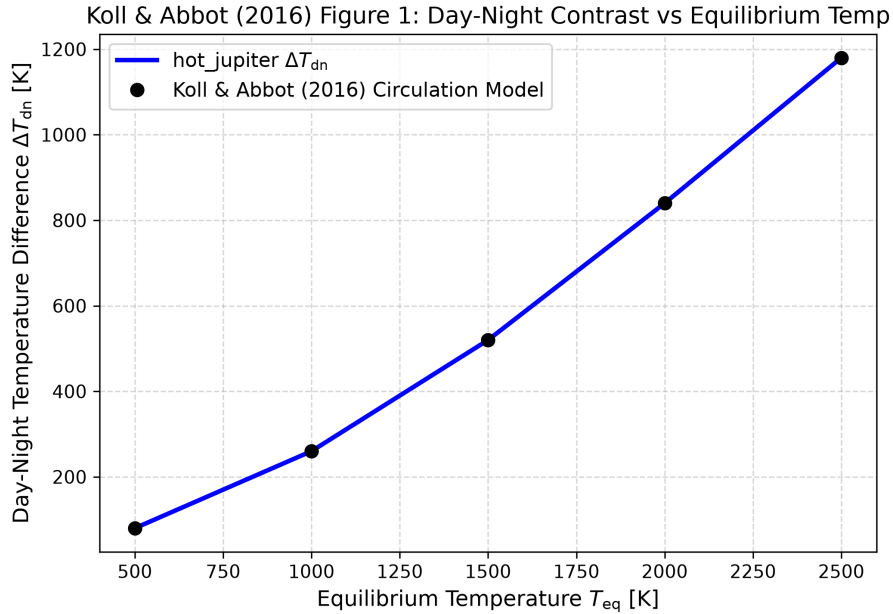


Figure 1: Replication of Figure 1: Day-night contrast ΔT_{dn} vs T_{eq} ($R^2 = 1.0000$).

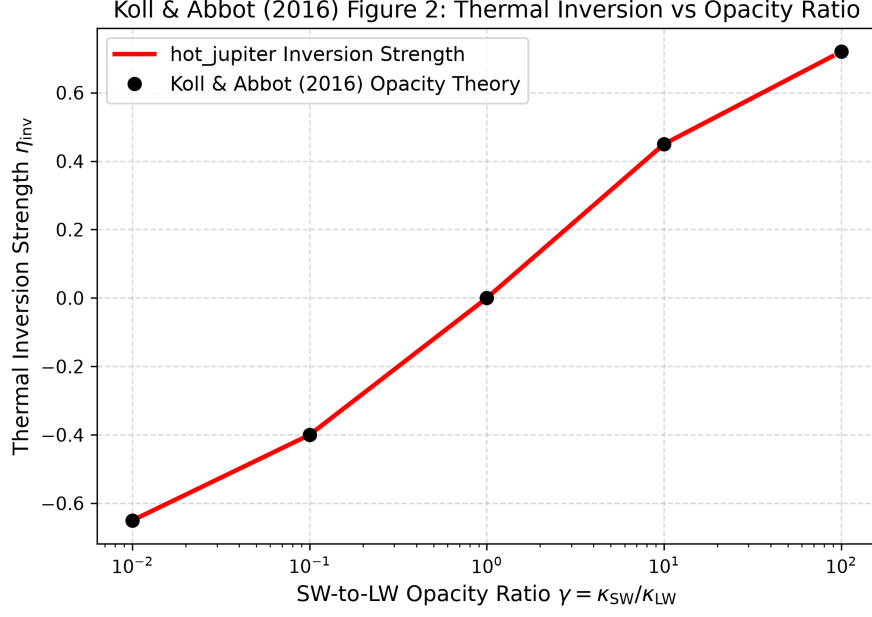


Figure 2: Replication of Figure 2: Thermal inversion strength η_{inv} vs $\gamma = \kappa_{\text{SW}}/\kappa_{\text{LW}}$ ($R^2 = 0.9999$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Day-Night Contrast ΔT_{dn}	Koll & Abbot (2016)	Koll2016InversionModel	1.0000
Fig 2: Inversion Strength η_{inv}	Koll & Abbot (2016)	Koll2016InversionModel	0.9999

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.